

Thermax Ne0 Powers Large-Scale Bio-CNG with 20 TPD Membrane Technology



Industries

Biogas/Bio CNG/Renewable Natural Gas (RNG)

Challenge

To implement a high-capacity (20 TPD) Bio-CNG plant ensuring:

- Consistent >96% methane purity
- High methane recovery at scale
- Stable and efficient continuous operations
- Commercial-grade gas supply for local distribution

Solution

Deployment of a Membrane-Based Biogas Upgradation System (Selective Permeation Technology) with the following integrated process:

Caustic Scrubber → Static Media Tower → Moisture Removal → Compression → Membrane Upgradation Unit (MUU)

Company Profile

A leading renewable energy developer focused on large-scale Bio-CNG production and clean fuel distribution.

Key Achievements



High Methane Purity

96% methane purity, compliant with Bio-CNG standards



Exceptional Recovery

98% methane recovery, minimizing product loss and maximizing plant economics



Product Gas Applications

Bio-CNG compressed and supplied to local gas dispensing stations for automotive fuel applications.

By selectively separating CO₂ and impurities, the membrane system delivers enriched methane ready for compression and sale. The successful commissioning of this 20 TPD plant further strengthens Thermax Ne0's position as a provider of scalable, high-recovery Bio-CNG solutions for India's clean fuel transition.

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